

Let $f(x) \in F[x]$ be given with the irreducible factorization

$$f(x) = f_1(x) \cdots f_n(x).$$

Since every Automorphism fixing F maps root of f_i to the root of f_i , the same irreducible Polynomial, therefore, for the splitting field K of f ,

$$\text{Aut}(K/F) \hookrightarrow S_{f_1} \times \cdots \times S_{f_n}.$$

With such an observation, in this paper, we figure out concretely the Galois group of the polynomials.

Def. Let $x_1, \dots, x_n$ be indeterminates.

The elementary symmetric functions $S_1, S_2, \dots, S_n$ are defined by

$$S_1 = x_1 + x_2 + \dots + x_n$$

$$S_2 = x_1 x_2 + x_1 x_3 + \dots + x_1 x_n + x_2 x_3 + x_2 x_4 + \dots + x_{n-1} x_n$$

$\vdots$

$$S_n = x_1 x_2 \dots x_n$$

And, the general polynomial of degree n is the polynomial

$$f(x) = (x - x_1)(x - x_2) \dots (x - x_n).$$

Note that

$$f(x) = x^n - S_1 x^{n-1} + S_2 x^{n-2} - \dots + (-1)^n S_n$$

Thm. Let $x_1, \dots, x_n$ be indeterminates of a field F , and $S_1, \dots, S_n$ be symmetric functions. Then, $F(x_1, \dots, x_n)$ is Galois extension of $F(S_1, \dots, S_n)$ with

$$\text{Gal}(F(x_1, \dots, x_n) / F(S_1, \dots, S_n)) \cong S_n.$$

Proof. Since $F(x_1, \dots, x_n)$ is the splitting field of the polynomial

$$f(x) = (x - x_1) \dots (x - x_n) = x^n - S_1 x^{n-1} + \dots + (-1)^n S_n$$

over $F(S_1, \dots, S_n)$ and f is separable, so the first assertion is obtained.

And, since degree of f is n , so $[F(x_1, \dots, x_n) : F(S_1, \dots, S_n)] \leq n!$.

Meanwhile, for any permutation $\sigma \in S_n$, a induced map

$$\bar{\sigma} : F(x_1, \dots, x_n) \rightarrow F(x_1, \dots, x_n) : \frac{g(x_1, \dots, x_n)}{f(x_1, \dots, x_n)} \mapsto \frac{g(x_{\sigma(1)}, \dots, x_{\sigma(n)})}{f(x_{\sigma(1)}, \dots, x_{\sigma(n)})}$$

is an Automorphism which fixes $F(S_1, \dots, S_n)$, so

$$S_n \hookrightarrow \text{Aut}(F(x_1, \dots, x_n) / F(S_1, \dots, S_n))$$

Since $|S_n| = n!$, so proved. @